Supplementary Results and Reliability Analyses

Supplementary record for `main.tex`

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1 Evaluation procedure

The guidance models were calibrated on the silver validation split and then evaluated once on the human-reviewed gold evaluation set without further adjustment. Reliability intervals use 2,000 non-parametric conversation-level bootstrap resamples with replacement and random seed 42. Sampling is not stratified. Guidance and end-to-end intervals are percentile 95% confidence intervals for macro-F1 over 200 resampled conversations. Topic macro-F1 uses the fixed set of 12 legal topics observed among the 84 guidance-positive gold conversations; end-to-end macro-F1 uses the 13 routing outcomes observed in the gold evaluation set.

Exact two-sided McNemar tests compare correctness on the same human-reviewed examples. Guidance and end-to-end comparisons use all 200 conversations, while topic-only comparisons use the 84 guidance-positive conversations.

2 Per-topic results

	Silver-labeled dataset			Gold evaluation set			
Topic	n	%	Any %	n	TF–IDF	MB-base	MB-large
Immigration & Citizenship	137	14.1	14.6	8	.800	.800	**.857**
Criminal Law	111	11.4	11.9	12	.857	**.870**	.846
Intellectual Property	110	11.3	12.0	7	.588	.875	**.933**
Business & Commercial	107	11.0	17.9	11	.640	.588	**.667**
Tax Law	105	10.8	13.3	12	.556	**.783**	**.783**
Employment & Labor	87	9.0	10.0	10	.889	**.900**	**.900**
Housing & Property	81	8.3	9.5	8	.857	**.933**	.800
Government & Administrative	73	7.5	15.2	5	.182	.000	**.444**
Consumer & Personal Finance	54	5.6	8.2	3	**.667**	.500	.500
Family & Estates	51	5.2	6.6	3	.667	**.750**	.571
Data Privacy & Technology	36	3.7	5.8	4	.000	**.750**	**.750**
Torts & Civil Liability	12	1.2	2.4	–	–	–	–
Other	5	0.5	0.6	–	–	–	–
Civil Rights & Constitutional	3	0.3	2.2	1	.000	.000	.000

Table 1: Legal topics by frequency in the silver-labeled dataset and gold per-class performance. Silver columns cover the 972 guidance-positive conversations, where n and % are primary-topic assignments and Any % also counts secondary labels; gold columns give support among the 84 human-reviewed guidance-positive conversations and per-class F1. The 2 classes without gold support are marked as unavailable and excluded from macro-F1, and five classes have five or fewer gold examples.

Performance varies substantially across topics. ModernBERT-large scores above .84 on "Intellectual Property," "Employment & Labor," "Immigration & Citizenship," and "Criminal Law," but .444 on "Government & Administrative" and .000 on "Civil Rights & Constitutional," which has one gold example. Five observed topics have five examples or fewer, so the per-class results are descriptive.

3 Topic and end-to-end confidence intervals

	Topic classification		End-to-end routing	
Model	Macro-F1	95% CI	Macro-F1	95% CI
TF–IDF + logistic regression	.559	[.441, .632]	.527	[.418, .591]
ModernBERT-base	.646	[.537, .715]	.544	[.450, .608]
ModernBERT-large	.671	[.538, .750]	.593	[.487, .667]

Table 2: Macro-F1 and 95% bootstrap confidence intervals for topic classification and end-to-end routing on the gold evaluation set. Topic classification uses gold guidance labels for the 84 guidance-positive conversations and the fixed set of 12 observed topics. End-to-end routing uses all 200 conversations and the 13 observed routing outcomes. Intervals use 2,000 conversation-level bootstrap resamples with replacement and seed 42.

Table 2 reports the topic and end-to-end macro-F1 intervals from the frozen evaluation bundle. The intervals overlap across the three systems; paired significance tests are reported below.

4 Full guidance-detection metrics

Model	Acc.	Macro-F1	95% CI	Prec.	Rec.	Pos. F1	AUPRC
Full conversation							
TF–IDF + logistic regression	.840	.839	[.784, .888]	**.765**	.893	.824	.866
ModernBERT-base	.815	.813	[.755, .865]	.747	.845	.793	.894
ModernBERT-large	**.845**	**.845**	[.793, .894]	.752	**.940**	**.836**	**.901**
Prefix models (final turn)							
TF–IDF + logistic regression	.825	.823	[.767, .873]	.763	.845	.802	.864
ModernBERT-base	.830	.828	[.771, .880]	.760	.869	.811	.903
ModernBERT-large	**.860**	**.859**	[.810, .905]	**.792**	**.905**	**.844**	**.940**

Table 3: Full guidance-detection metrics on the 200-conversation gold evaluation set. Precision, recall, and positive F1 are for the guidance class. Macro-F1 intervals use 2,000 conversation-level bootstrap resamples with replacement and seed 42. Prefix models are scored after the final user turn. Bold marks the best value within each block.

5 Exploratory early-prefix results

	Turn 1 ($n = 200$)		Turn 2 ($n = 104$)	
Model	Acc.	Macro-F1	Acc.	Macro-F1
TF–IDF + logistic regression	**.830**	**.826**	.798	.798
ModernBERT-base	.800	.792	.808	.808
ModernBERT-large	.825	.819	**.865**	**.865**

Table 4: Exploratory initial-turn prediction results. Turn 1 uses all conversations in the gold evaluation set; Turn 2 uses the 104 conversations containing at least two user turns. Labels indicate whether the complete conversation eventually seeks guidance, not whether guidance has begun by that turn. The columns use different evaluation sets and should not be interpreted as within-model improvement.

Turn-1 bootstrap 95% intervals for macro-F1 overlap: TF–IDF [.770, .876], ModernBERT-base [.730, .846], and ModernBERT-large [.766, .873]. Among the 84 gold-positive conversations, the turn-1 models identify 67, 60, and 65 cases, respectively. Turn 2 is restricted to 53 gold-positive conversations with at least two user turns; TF–IDF identifies 43, ModernBERT-base 41, and ModernBERT-large 44. These results use eventual conversation-level guidance labels rather than human-reviewed onset labels.

6 Paired significance tests

Task	Comparison (A vs. B)	A only	B only	p
Guidance	ModernBERT-large vs. TF–IDF + LogReg	14	13	1.000
	ModernBERT-large vs. ModernBERT-base	15	9	.307
	ModernBERT-large (prefix) vs. ModernBERT-large	12	9	.664
	ModernBERT-large (prefix) vs. ModernBERT-base	16	7	.093
Topic only	ModernBERT-large vs. TF–IDF + LogReg	13	6	.167
	ModernBERT-large vs. ModernBERT-base	5	4	1.000
Cascade	ModernBERT-large vs. TF–IDF + LogReg	22	19	.755
	ModernBERT-large vs. ModernBERT-base	18	12	.362

Table 5: Two-sided exact paired McNemar tests on the gold evaluation set. "A only" and "B only" count conversations exactly one system classified correctly; the test uses only these discordant pairs. Guidance and end-to-end tests use all 200 conversations, while topic-only tests use the 84 guidance-positive ones. No comparison reaches $p < .05$.

None of the eight comparisons reaches $p < .05$. These tests provide no statistically significant evidence that one evaluated system is superior to another. ModernBERT-large has the highest numerical full-conversation guidance macro-F1 (.845) and end-to-end macro-F1 (.593).

Human review changed the silver guidance decision for 8 of the 200 gold conversations. Across the six guidance detectors, accuracy on these cases ranges from .375 to .500, compared with .833 to .880 on the other 192. These cases may be more difficult, and the group is too small for a stable estimate. Among the 7 gold guidance-positive conversations whose silver primary topic changed, ModernBERT-base and ModernBERT-large each predict the final topic correctly in 5 cases, while TF–IDF does so in 3.

7 Resources

The Legal Guidance in the Wild Hugging Face collection provides the datasets, checkpoints, training details, and evaluation artifacts. The public evaluation script, `scripts/evaluate_gold.py`, recomputes the gold-set results from the frozen predictions in `supplementary/evaluation/`.